Politecnico di Milano
School of Industrial and Information Engineering

Atom Optics and Quantum Technologies
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Part 1 – Fluctuations and heating within an Optical Molasses.

Consider a cloud of atoms within an optical molasses. The quantum fluctuations of the Resonance Radiation Pressure (RRP) force can be modeled by the expression $\mathcal{F}(t) = F_{RRP} + f(t)$, where F_{RRP} is the average (ensemble) RRP force and $f(t)$ is the so-called Langevin force.

a) Explain how the mathematical modeling includes the absorption and scattering contributions $\mathcal{F}_{abs}$ and $\mathcal{F}_{scat}$, respectively. In particular, provide expressions for $\mathcal{F}_{abs}$ and $\mathcal{F}_{scat}$ and discuss their main features.
b) Consider the simplified case of fluctuations with a single laser beam. Derive the rate of change of the variance of the atomic linear momentum due to absorption and scattering (heating contributions).[Suggestion: $\left.\frac{d\sigma_{\mathrm{p}}^2}{dt}\right|_{\mathrm{abs}} = \hbar^2 \mathrm{k}^2 \Gamma_{\mathrm{p}}$, $\left.\frac{d\sigma_{\mathrm{p}}^2}{dt}\right|_{\mathrm{scat}} = \hbar^2 \mathrm{k}^2 \Gamma_{\mathrm{p}}$].
c) Explain how this model extends to the case of 3D optical molasses.
d) Derive the rate of change of the variance of the atomic linear momentum due to the damping force F_{OM} exerted by the optical molasses (cooling contribution). [Suggestion: $\left.\frac{d\sigma_{\mathrm{p_i}}^2}{dt}\right|_{\mathrm{cooling}} = -2\frac{\alpha}{\mathrm{M}}\sigma_{\mathrm{p_i}}^2$].
e) Derive the expression for the Doppler cooling limit temperature T_{D}. To this purpose, the expression of the damping coefficient α is given by:

$$\alpha = -\frac{8\hbar \mathrm{k}^2 \mathrm{s}_0}{\Gamma} \frac{\Delta\omega}{\left[1 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}$$

f) Explain the physical mechanism behind the interplay of heating and cooling in an optical molasses at equilibrium. Support your explanation with a sketch of the velocity trajectory behavior.

Part 2 – Light Shift

The Light Shift for a transition from state a to b with polarization ε is $\widetilde{\mathrm{E}}_{\pm} = \pm\frac{\hbar\Delta\omega}{2} \pm \frac{\hbar\Omega_{ab,\varepsilon}{}^2}{4\Delta\omega^2}$, where $\Omega \ll \Delta\omega$ is assumed. Explain the physical meaning of this effect using a graphical representation in both the bare and dressed state pictures. Explain the link between the Rabi frequency $\Omega_{ab,\varepsilon}$ and the line strength $\mathrm{S}_{ab,\varepsilon}$.

Part 3 – Polarization gradient cooling (Sisyphus cooling)

Explain the theory behind the so-called Sisyphus cooling. In this case, derive the limit temperature T_{Sisyphus} and show how it is related to the light shift effect.